Secret-Learn — 六层架构 · 隐私计算数据流

sklearn-compatible API → privacy mode → JAX / SecretFlow compute → secure devices

Primary flow
Fallback / feedback
Dashed = architecture zone
Box = module + responsibility

PROJECT OVERVIEW Secret-Learn exposes sklearn-compatible machine learning across federated, secret-sharing, and split-learning modes.
It combines JAX acceleration with SecretFlow devices so each workflow can choose where plaintext, shares, and activations reside.

APPLICATION · 多方隐私数据

- **Healthcare / Finance** — cross-party modeling
- **IoT / Research** — distributed learning

SKLEARN API · 统一算法入口

- **secretlearn.*** — fit · predict · transform
- **algorithm_classifier.py** — classify fit signature / epochs

PRIVACY MODES · execution placement

- **FEDERATED LEARNING** — PYU local compute; HEU secure aggregation
- **SECRET SHARING** — SPU MPC · ABY3 / CHEETAH
- **SPLIT LEARNING** — split model · encrypted activations

ACCELERATION & COMPUTE

- **JAX Acceleration** — JIT · vmap · xlearn
- **SecretFlow Runtime** — party orchestration

SECURE DEVICES

- **PYU** — party-local plaintext
- **HEU** — homomorphic aggregation
- **SPU** — secure MPC
- **TEE** — optional enclave

INFRASTRUCTURE & HARDWARE

- **SecretFlow** — distributed secure runtime
- **JAX / XLA** — CPU · GPU · TPU
- **sklearn** — API compatibility

同一 sklearn 调用由 FL / SS / SL 决定数据驻留与执行设备；JAX 加速可公开的本地计算。